Generating Consistent LLM-Based NPC Dialogue

by Structuring and Prioritising Game Context

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Bachelor of Advanced Computing (Honours)



Motivation and Context

LLM Application for generating dialogue

- ChatGPT: Generates fluent text for open conversations
- Character.AI: Supports real-time chats with fictional personas
- NovelAI: Helps writers generate creative story dialogue
- Neuro-sama: AI streamer that respond to viewer in real-time









Motivation and Context

LLM Difficulty in game dialogue generation



From Article: R. Rabbit, "AI Powered NPCs — Hype, or Hallucination? - curiouserinstitute - Medium," *Medium*, Dec. 09, 2023. https://medium.com/curiouserinstitute/ai-powered-npcs-hype-or-hallucination-11ddfc530e33

Motivation and Context

LLM Difficulty in game dialogue generation

Game worlds have their own facts

Motivation and Context

Game worlds have their own facts

how to make a door

🪵 Basic Wooden Door (Simple Method)

Materials:

- Wooden boards or a plywood sheet (for the door panel)
- 2–3 horizontal support pieces (called rails)
- Screws or nails
- Wood glue (optional but stronger)
- Hinges (2–3)
- Door handle or knob
- Sandpaper
- Saw, drill, measuring tape

🔪 Steps:

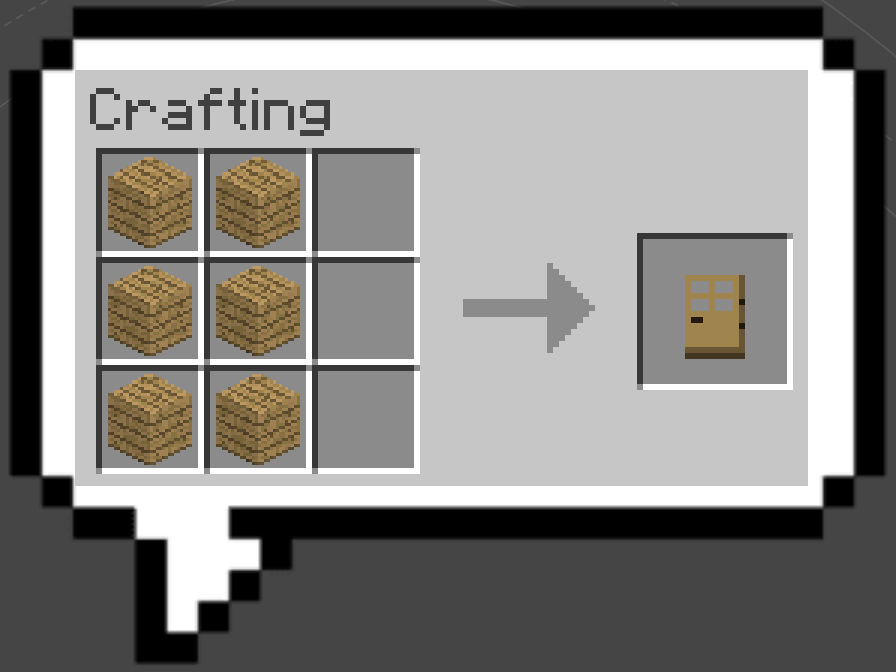
1. Measure the Door Frame

- Measure the **height and width** of the opening.
- Subtract about **5–10 mm (¼–½ inch)** so the door can open easily.

Motivation and Context

Game worlds have their own facts





how to make a door

Motivation and Context

LLM Difficulty in game dialogue generation

NPC should know only certain information

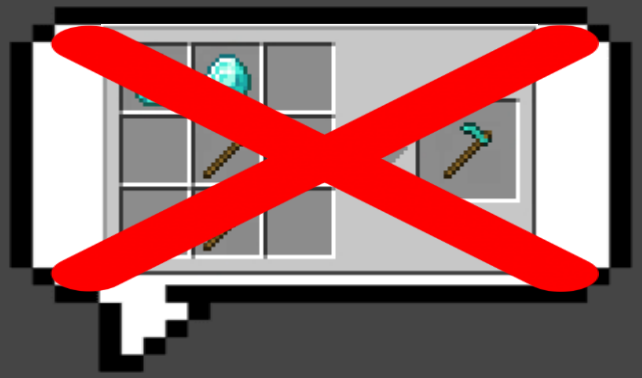
Game worlds have their own facts

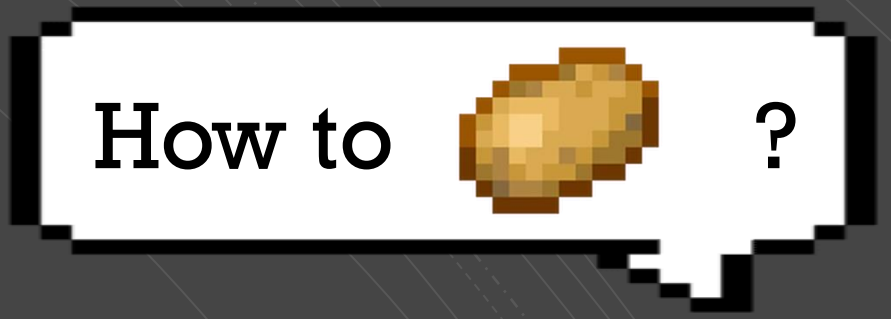
Motivation and Context

NPC should know only certain information









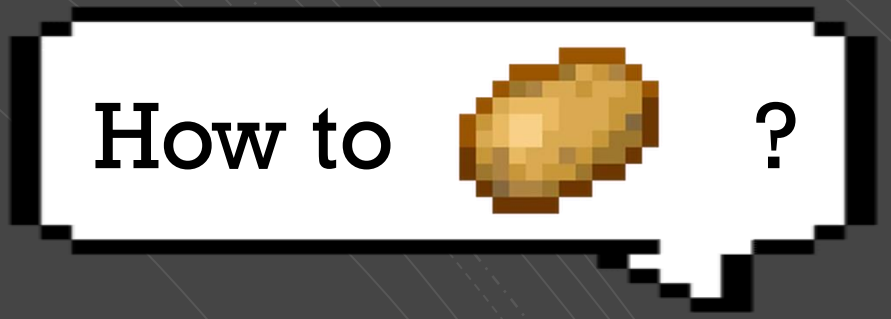
Motivation and Context

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Motivation and Context

LLM Difficulty in game dialogue generation

NPC should know only certain information

Game worlds have their own facts

Dialogue must match possible actions in game

Motivation and Context

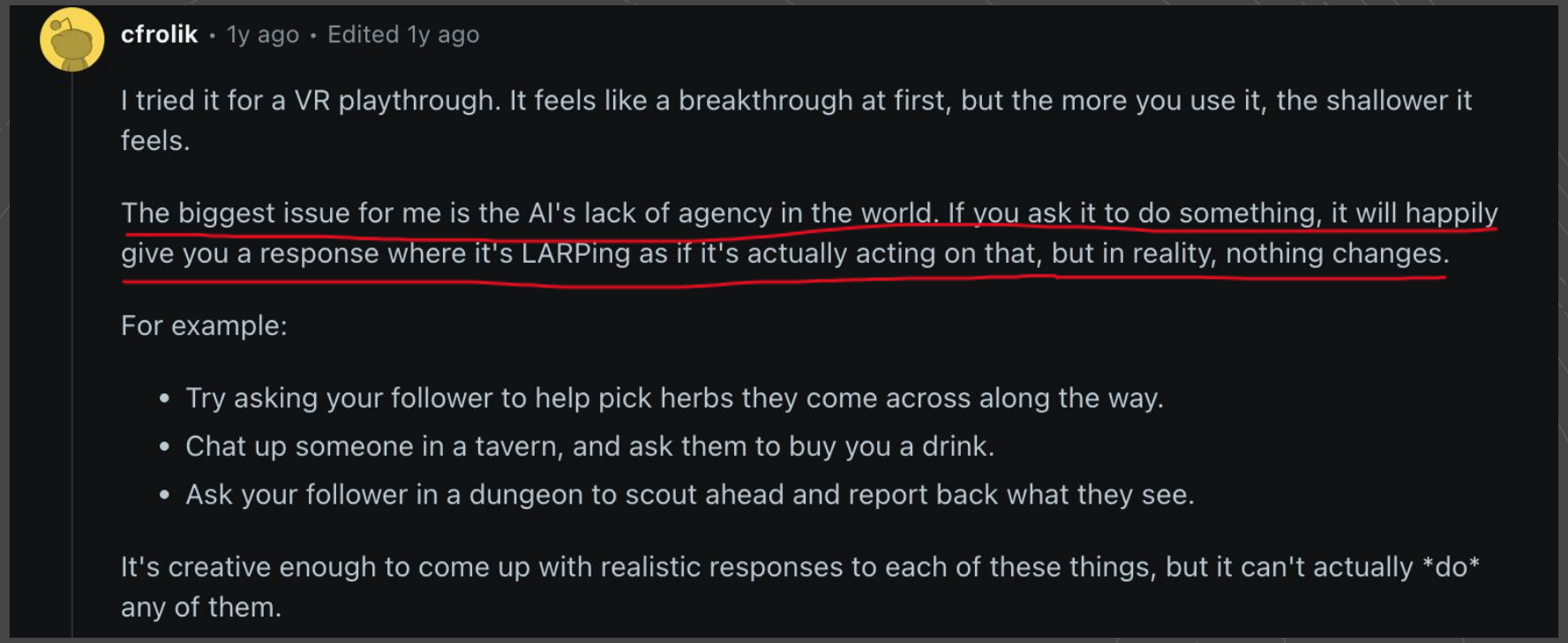
Dialogue must match possible actions in game



From mod page: "Herika - The ChatGPT Companion," Nexus Mods :: Skyrim Special Edition, Sep. 08, 2023. https://www.nexusmods.com/skyrimspecialedition/mods/89931

Motivation and Context

Dialogue must match possible actions in game



cfrolik • 1y ago • Edited 1y ago

I tried it for a VR playthrough. It feels like a breakthrough at first, but the more you use it, the shallower it feels.

The biggest issue for me is the AI's lack of agency in the world. If you ask it to do something, it will happily give you a response where it's LARPing as if it's actually acting on that, but in reality, nothing changes.

For example:

- Try asking your follower to help pick herbs they come across along the way.
- Chat up someone in a tavern, and ask them to buy you a drink.
- Ask your follower in a dungeon to scout ahead and report back what they see.

It's creative enough to come up with realistic responses to each of these things, but it can't actually *do* any of them.

From reddit post: Reddit.com, 2026.”Ok..CHIM or Herika AI, is INCREDIBLE !”. https://www.reddit.com/r/skyrimvr/comments/1h5z0k2/okchim_or_herika_ai_is_incredible/ (accessed May 10, 2026).

Critical Literature Review

keywords

- LLM (Large Language Model)
- AI NPC
- **Context-aware** (context representation and structuring)
- **Knowledge-grounded**, world context, and filtering
- **Memory model** for store and retrieval information

Critical Literature Review

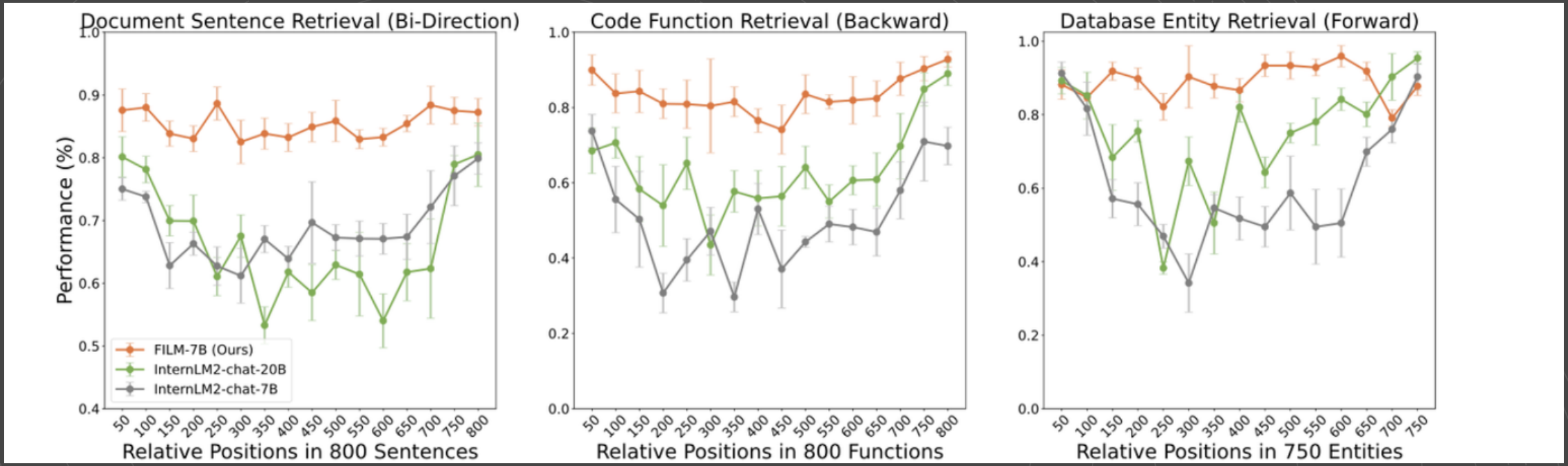
LLM-based NPC dialogue system



From paper: Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. In Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology.

Critical Literature Review

Limitation of Unified Context Pipeline



From paper: An, S., Ma, Z., Lin, Z., Zheng, N., Lou, J.-G., & Chen, W. (2024). Make Your LLM Fully Utilize the Context. In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS 2024).

Critical Literature Review

LLM Structured Context Pipeline

Paper	Context Used
Generative Agent (2023)	current observation + retrieved memories + reflections + plans
Character-LLM (2023)	character profile => build protective character's experience
CharacterCraft (2025)	profiles + scenes + emotions + topics + past interactions.
CharacterBox (2025)	scenes + character state + interactions + narrator updates + changes
Beyond Dialogue (2025)	role profiles + scenario-specific dialogue
MemoryRepository (2024)	Separates short-term dialogue from summarised long-term memory
…	…

Critical Literature Review

Gap in knowledge

- Context choices are rarely justified
- Context influence are not clearly explained
- Context organisation has limited evidence
- Few studies ablate context influence directly
- Therefore, it's unclear whether a prioritised structured context pipeline would improve NPC dialogue consistency

Research Question

Research Objective

How do different strategies for **structuring and prioritising** game-specific context affect the **consistency and realism** of AI-generated dialogue for game characters?

Research Question

Hypothesis

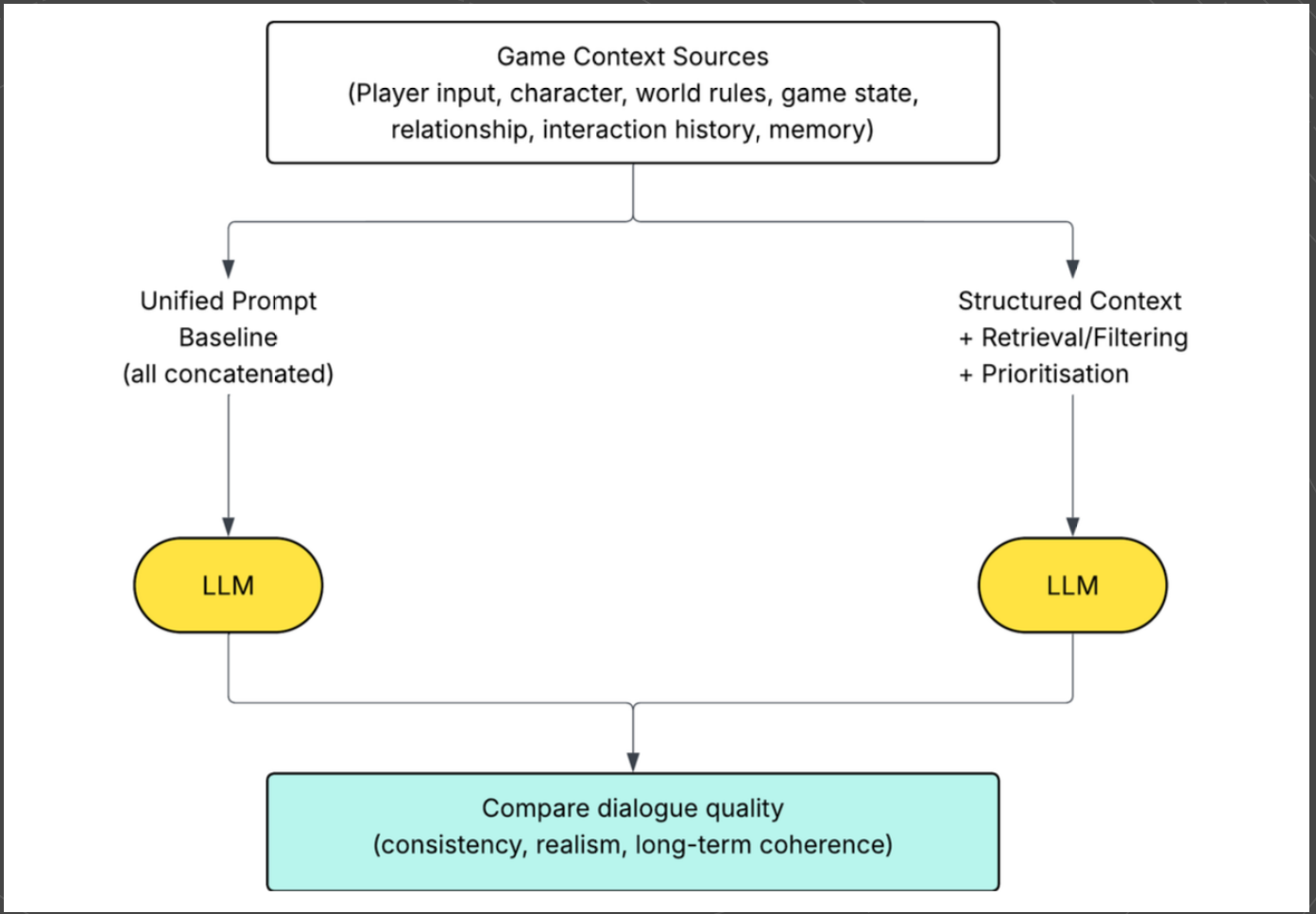
H1: NPC dialogue generated with a prioritised structured context pipeline will achieve higher **consistency** than dialogue generated with a unified context prompt.

H2: NPC dialogue generated with a prioritised structured context pipeline will achieve higher **realism** than dialogue generated with a unified context prompt.

H3: NPC dialogue generated with a prioritised structured context pipeline will maintain higher **long-term consistency across turns** than dialogue generated with a unified context prompt.

Research Methods

Planned Contribution



Research Methods

Variables

Independent Variable: The way contexts are structured and prioritised

Dependent Variable: Consistency and Realism (must be measurable)

Control Variable

- LLM model
- World settings
- NPC profile
- Current scenario
- User prompt
- Number of dialogue turn (for H3)

Research Methods

Experimental Procedure

1. Scenario preparation (Design World, NPC, etc.)
2. Apply context condition (unified vs prioritised context structure)
3. Generate NPC responses (same prompt, different context structure)
4. Evaluate response (using evaluation metrics)
5. Aggregate results (across turn for H3)
6. Compare conditions (finalise and prove hypothesis)

Research Methods

Evaluation Metrics

H1:

$$\text{Consistency} = \frac{\text{Recall} + \text{Correctness} + \text{World Consistency} + \text{No Contradiction}}{4}$$

H2:

$$\text{Realism} = \frac{\text{Coherence} + \text{Context Use} + \text{Relevance} + \text{Naturalness}}{4}$$

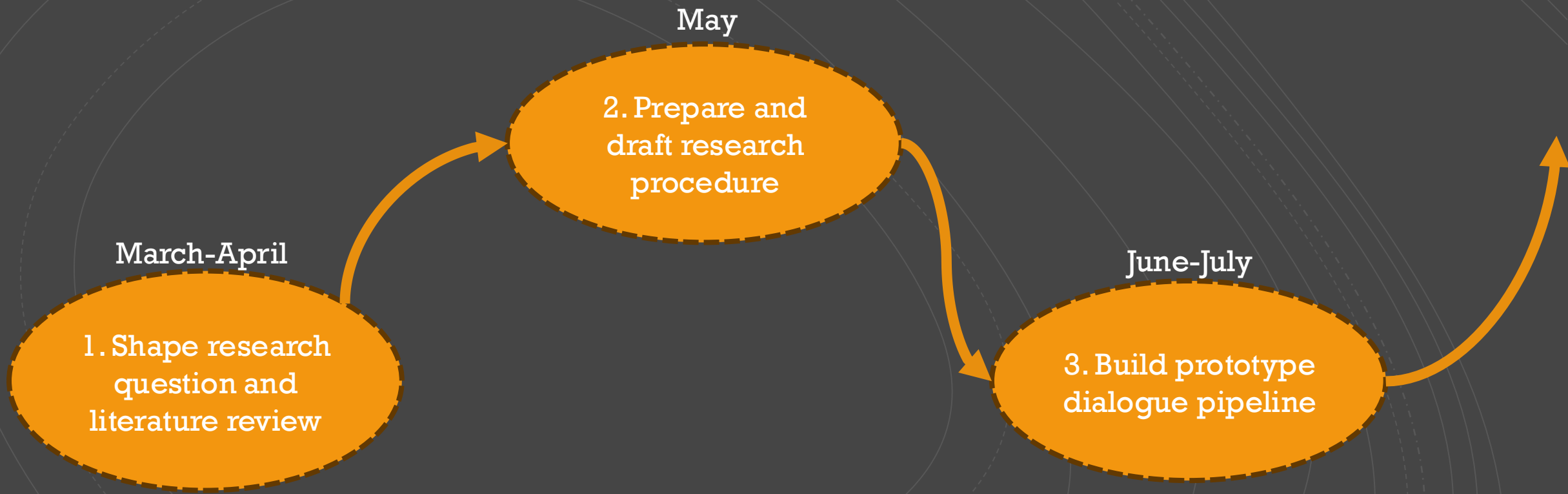
H3:

$$\textbf{Long-term Consistency} = \frac{\text{Consistency}_1 + \text{Consistency}_2 + \cdots + \text{Consistency}_T}{T}$$

$$\textbf{Decline Rate} = \frac{1}{T-1} \sum_{t=1} (\text{Consistency}_t - \text{Consistency}_{t+1})$$

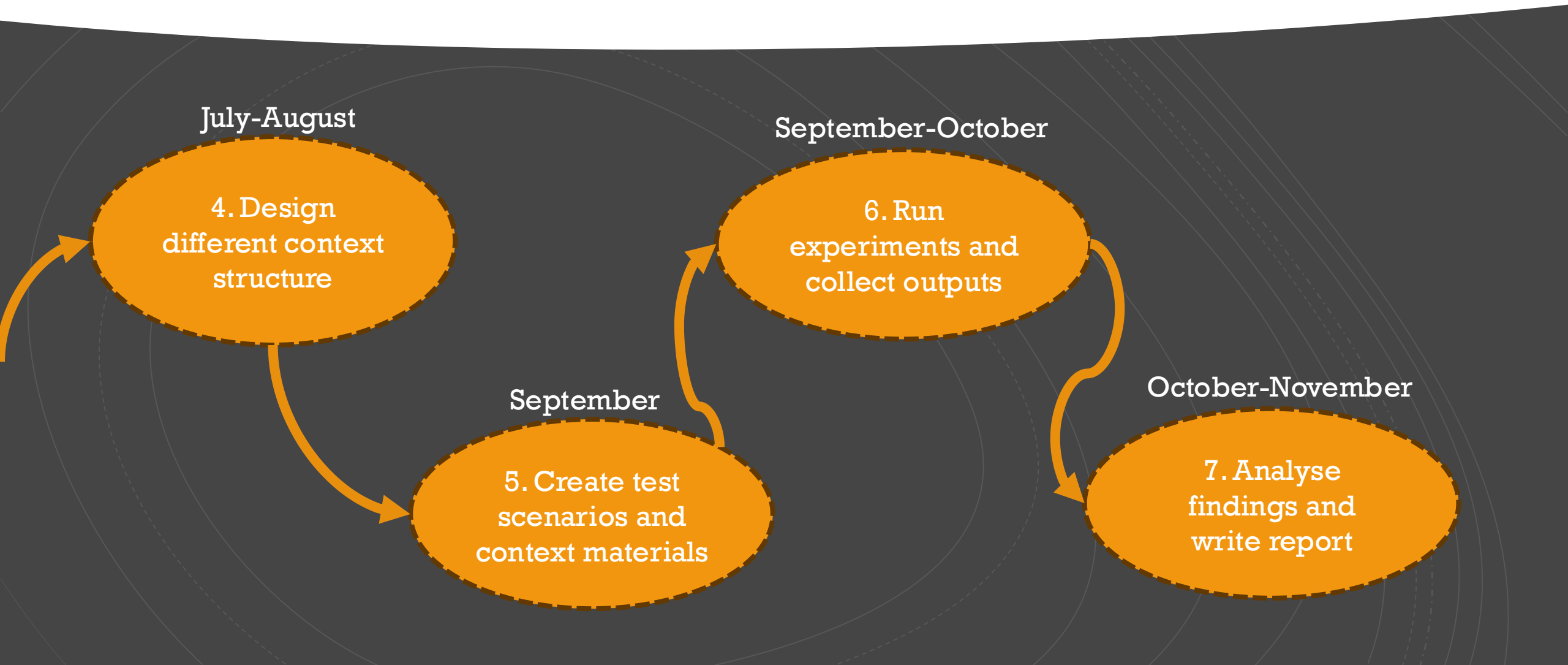
Plan

Timeline



Plan

Timeline



Summaries

- LLM NPCs are fluent, but often inconsistent with the game world
- Prior work adds context, but rarely prioritises or manage it.
- This research tests whether prioritised context improves dialogue.

References

- Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. In Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology.
- An, S., Ma, Z., Lin, Z., Zheng, N., Lou, J.-G., & Chen, W. (2024). Make Your LLM Fully Utilize the Context. In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS 2024).
- Shao, Y., Li, L., Dai, J., & Qiu, X. (2023). Character-LLM: A trainable agent for role-playing. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing.
- Yang, B., Liu, D., Xiao, C., Zhao, K., Tang, C., Li, C., Yuan, L., Guang, Y., & Lin, C. (2025). Crafting customisable characters with LLMs: A persona-driven role-playing agent framework. In Findings of the Association for Computational Linguistics: EMNLP 2025.
- Wang, L., Lian, J., Huang, Y., Dai, Y., Li, H., Chen, X., Xie, X., & Wen, J.-R. (2025). CharacterBox: Evaluating the role-playing capabilities of LLMs in text-based virtual worlds. In Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies.
- Yu, Y., Yu, R., Wei, H., Zhang, Z., & Qian, Q. (2025). Beyond dialogue: A profile-dialogue alignment framework towards general role-playing language model. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics.
- Zheng, S., He, K., Yang, L., & Xiong, J. (2024). MemoryRepository for AI NPC. IEEE Access, 12, 62581–62596